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Process chamber

Abstract

A process chamber is provided including a chamber body enclosing an interior volume and a substrate support assembly in the interior volume. The substrate support assembly includes a substrate support having a substrate receiving surface, an upper plate, and a lower plate, the upper plate positioned over the substrate support. The substrate support is positioned over and spaced apart from the lower plate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8216375	12/2011	Liu	N/A	N/A
10281261	12/2018	Paul et al.	N/A	N/A
10544519	12/2019	Savas et al.	N/A	N/A
10861722	12/2019	Colombeau et al.	N/A	N/A
2003/0049372	12/2002	Cook	118/724	C23C 16/345
2004/0053515	12/2003	Comita	257/E21.119	C30B 33/00
2004/0144323	12/2003	Kai	N/A	N/A
2011/0027973	12/2010	Su	257/E21.09	C23C 16/481
2012/0263875	12/2011	Brenninger et al.	N/A	N/A
2014/0079526	12/2013	Oyama	269/55	H01L 21/68
2014/0261159	12/2013	Okabe et al.	N/A	N/A
2014/0326185	12/2013	Lau	118/728	C23C 16/45563
2017/0148654	12/2016	Paul et al.	N/A	N/A
2018/0182652	12/2017	Seshimo	N/A	H01L 21/67017
2021/0351014	12/2020	Noorbakhsh	N/A	H01J 37/32513
2022/0064790	12/2021	Wada	N/A	H01L 21/67766
2022/0325400	12/2021	Cong et al.	N/A	N/A

OTHER PUBLICATIONS

Matweb.com. Silicon properties. cited by examiner

Matweb.com. Silicon carbide properties. cited by examiner

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Background/Summary

BACKGROUND

(1) Embodiments of the present disclosure generally relate to improved process chambers, such as improved semiconductor process chambers configured to perform depositions and cleaning procedures.

DESCRIPTION OF THE RELATED ART

(2) Deposition process chambers are cleaned frequently to improve the process results of the depositions performed in the process chambers. For some depositions, the interior of the process

chamber is cleaned after each deposition or after a small number of depositions, such as less than ten depositions. After a specified number of depositions is performed in the chamber, the substrate from the last deposition is removed and the process chamber is cleaned. After the cleaning is complete, a new substrate is moved into the chamber for the next deposition. Each cycle of cleaning has an effect on the throughput of the process chamber and the costs of production for the product being formed.

(3) Accordingly, there is a need for improved process chambers and related methods for increasing the throughput of process chambers and reducing the downtime relating to the cleaning of the process chambers.

SUMMARY

(4) In one embodiment, a process chamber is provided including a chamber body enclosing an interior volume; and a substrate support assembly in the interior volume, the substrate support assembly including a substrate support having a substrate receiving surface, an upper plate, and a lower plate, the upper plate positioned over the substrate support, wherein the substrate support is positioned over and spaced apart from the lower plate.

(5) In another embodiment, a process chamber is provided including: a chamber body enclosing an interior volume; a substrate support assembly in the interior volume, the substrate support assembly including a substrate support having a substrate receiving surface, an upper plate, and a lower plate, the upper plate positioned over the substrate support, wherein the substrate support is positioned over and spaced apart from the lower plate; and a first ring positioned above the lower plate of the substrate support assembly, wherein a portion of the upper plate of the substrate support assembly extends over the first ring.

(6) In another embodiment, a method of processing a substrate is provided including introducing one or more cleaning gases to a first portion of an interior volume of a process chamber located above a substrate support assembly disposed in the interior volume of the process chamber, and cooling a substrate on a substrate support of the substrate support assembly during the introducing of the one or more cleaning gases to the first portion of the interior volume, wherein the substrate support is located in a second portion of the interior volume that is below the first portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

(2) FIG. 1A is a cross-sectional view of a processing system, according to one embodiment.

(3) FIG. 1B is a close-up cross-sectional view of section 1B of FIG. 1A showing additional detail of the substrate support assembly and liner assemblies, according to one embodiment.

(4) FIG. 1C is a cross-sectional view of the processing system of FIG. 1A with the substrate support assembly in a cleaning position, according to one embodiment.

(5) FIG. 2A is a cross-sectional view of another processing system with a substrate support assembly in a process position, according to one embodiment.

(6) FIG. 2B is a cross-sectional view of the processing system of FIG. 2A with the substrate support assembly in a cleaning position, according to one embodiment.

(7) FIG. 3A is a cross-sectional view of another processing system with a substrate support assembly in a process position, according to one embodiment.

- (8) FIG. 3B is a cross-sectional view of the processing system of FIG. 3A with the substrate support assembly in an intermediate position, according to one embodiment.
- (9) FIG. 3C is a cross-sectional view of the processing system of FIG. 3A with the substrate support assembly in a cleaning position, according to one embodiment.
- (10) FIG. 4 is a process flow diagram of a method of processing a substrate in a process chamber and cleaning the process chamber, according to one embodiment.
- (11) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(12) Embodiments of the present disclosure generally relate to improved process chambers, such as improved semiconductor process chambers configured to perform depositions and cleaning procedures. The process chambers described in this disclosure include substrate support assemblies that are configured to move to different positions inside the process chamber and substantially seal different regions of the interior volume of the process chamber from each other in these different positions. The substrate support assembly can include an upper plate and a lower plate. The upper plate and/or the lower plate can be positioned against or spaced apart from another component (e.g., a liner) by a small distance in the different positions, so that the different regions of the interior volume can be substantially sealed from each other. Sealing the different regions of the interior volume from each other enables the different regions to be operated under significantly different conditions. For example, a recently processed substrate can be cooled in a first portion of the process chamber while a high-temperature cleaning process can be performed in a different region of the process chamber. The seal between these two different regions can be used to prevent the highly active cleaning gases from interacting with the substrate that is cooling. Furthermore, a processed substrate in the cooling region of the process chamber can be exchanged with a new substrate for the next process (e.g., deposition) while the cleaning procedure is being executed in the cleaning region of the process chamber. Enabling the substrate to cool and be exchanged while a cleaning procedure is performed in a different region of the interior volume can significantly increase throughput for a process chamber.

(13) FIG. 1A is a cross-sectional view of a processing system **100**, according to one embodiment. The processing system **100** includes a process chamber **101**, a process gas source **121**, a cleaning gas source **122**, a purge gas source **124**, a chilled gas source **126**, an exhaust pump **129**, and a controller **175**. The processing system **100** can be configured to perform epitaxial deposition processes in the process chamber **101**. Although the disclosure mainly describes improved processing systems for increasing throughput for epitaxial deposition chambers, the improvements can be applied to other systems for processing substrates in which the substrate is conventionally removed before cleaning the interior of chamber used in the processing system.

(14) The process chamber **101** includes a chamber body **102**. The chamber body **102** includes an upper body **103**, a lower body **104**, a lid **106** positioned on the upper body **103**, a gas inlet module **107**, and a gas outlet module **108**. The lower body **104** can form a bottom **105** of the chamber body **102**. The chamber body **102** encloses an internal chamber **110** formed of an infrared transparent material (e.g., quartz). The internal chamber **110** includes an upper chamber **111** and a lower chamber **112** each formed of an infrared transparent material (e.g., quartz). The internal chamber **110** at least partially encloses an interior volume **115** of the process chamber **101**.

(15) The process chamber **101** includes an upper lamp module **116** and a lower lamp module **117**. The upper lamp module **116** is located between the lid **106** and the upper chamber **111**. The lower lamp module **117** is located between the bottom **105** of the lower body **104** and the lower chamber **112**. The upper lamp module **116** includes a plurality of lamps **116L** configured to direct heat through the upper chamber **111** and into the interior volume **115** during processing. The lower lamp

module **117** includes a plurality of lamps **117L** configured to direct heat through the lower chamber **112** and into the interior volume **115** during processing. The lower lamp module **117** can include a floor **118**. The plurality of lamps **117L** can be supported by the floor **118**.

(16) The process chamber **101** includes a substrate support apparatus **145**. The substrate support apparatus **145** includes a shaft **146**, an actuator **147**, and a substrate support assembly **150**. The shaft **146** couples the actuator **147** to the substrate support assembly **150**. The actuator **147** can be configured to raise and lower the substrate support assembly **150** by moving the shaft **146** in a vertical direction. The actuator **147** can also be configured to rotate the shaft **146** and the substrate support assembly **150** about a central vertical axis C extending through a center of the shaft **146**.

(17) The substrate support assembly **150** includes a lower plate **153** connected (e.g., fastened) to the shaft **146**. The substrate support assembly **150** includes a plurality of inner supports **155** (e.g., three pins) positioned on the lower plate **153**. The substrate support assembly **150** includes a susceptor **156** (also referred to as substrate support) positioned on the plurality of inner supports **155**. A substrate **50** is positioned on the susceptor **156**. The central vertical axis C can also extend through a center of the lower plate **153** and a center of the susceptor **156**.

(18) The substrate support assembly **150** includes a support ring **160** positioned on the lower plate **153**. The support ring **160** includes a lower portion **161**, an upper portion **162**, and a central ringed portion **163**. The central ringed portion **163** can form a complete ring (i.e., 360 degrees), for example around the central vertical axis C. The upper portion **161** and the lower portion **162** can be shaped like pins, and in some embodiments the portions **161**, **162** only extend a few degrees around the central vertical axis C. The pin shape of the portions **161**, **162** allows for gas flow around the portions **161**, **162**, so that process gas can be provided over the substrate **50** and gases can be exhausted from regions over the substrate **50**. On the other hand, the ring shape of the central ringed portion **163** and the proximity of the central ring portion **163** to the preheat ring **135** and the susceptor **156** reduces gas flow from regions below the central ringed portion **163** to regions above the central ringed portion **163** and in the opposite direction as well. In some embodiments, the central ring portion **163** is positioned from about 0.1 mm to about 2.0 mm, such as from about 0.4 mm to about 1.0 mm from the preheat ring **135** to reduce the gas flow between the different regions.

(19) The support ring **160** is located at a further horizontal distance from the central vertical axis C than the plurality of inner supports **155** are located relative to central vertical axis C. The substrate support assembly **150** further includes an upper plate **168** positioned on the top of the support ring **160**. The upper plate **168** and the lower plate **153** each have a larger cross-sectional area in a horizontal plane than a cross-sectional area of the susceptor **156** in a horizontal plane. In some embodiments, the inner supports **155**, the support ring **160**, the lower plate **153**, and the upper plate **168** can be formed of quartz. In some of these embodiments, these components can be formed of a low-OH quartz material having an OH content of less than about 30 parts per million (ppm), such as less than about 15 ppm, such as about 5 ppm or less. The low-OH content of the quartz can reduce the amount of radiation that is absorbed by these components.

(20) The process chamber **101** further includes an upper liner assembly **130** positioned around an outer portion of the interior volume **115**. In some embodiments, the upper liner assembly **130** can be attached to different portions of the chamber body **102** as well as to the upper chamber **111**. In some embodiments, the upper liner assembly **130** can extend 360 degrees around the interior volume **115** with the same shape or substantially the same shape for the entire 360 degrees. Additional details on the upper liner assembly **130** is provided in reference FIG. 1B described below. In some embodiments, the upper liner assembly **130** (also referred to as the hot-cavity liner) can be formed of quartz, such as silicon-impregnated quartz. The upper liner assembly **130** can be configured to have low thermal conductivity. For example, an upper liner assembly **130** formed of quartz can have a thermal conductivity of 1.38 W/m.K at 20° C.

(21) The upper liner assembly **130** can be positioned around the substrate support assembly **150**,

for example when the substrate support assembly **150** is in the raised processing position **150.sub.1** shown in FIG. **1A**. The process chamber **101** can further include a preheat ring **135** positioned on an inner ledge **134** of the upper liner assembly **130**.

(22) The substrate support assembly **150** and the upper liner assembly **130** can substantially seal a processing region **151** from a remainder of the interior volume **115** when the substrate support assembly **150** is in the processing position **150.sub.1**. The processing region **151** is bounded (1) on the upper side by the upper plate **168**, (2) on the lower side by the lower plate **153**, and (3) on the sides by the upper liner assembly **130**. The upper plate **168** can be positioned from about 0.1 mm to about 2.0 mm, such as from about 0.4 mm to about 1.0 mm from the upper liner assembly **130**, which can substantially seal the portion of the interior volume **115** above the upper plate **168** from the processing region **151**, which can reduce unintended depositions in this portion of the interior volume **115**. The lower plate **153** can be positioned from about 0.1 mm to about 2.0 mm, such as from about 0.4 mm to about 1.0 mm from a lower portion **132** of the upper liner assembly **130**, which can substantially seal the processing region **151** from portions of the interior volume **115** below the lower plate **153**.

(23) The process chamber **101** further includes a lower liner assembly **140** (also referred to as the cold-cavity liner) positioned below the upper liner assembly **130**. The lower liner assembly **140** can assist in cooling the temperature of the substrate **50** when the substrate **50** is separated from a cleaning environment as shown in FIG. **1C** described below. In some embodiments, the lower liner assembly **140** can be formed of a material having a higher thermal conductivity than the upper liner assembly **130**, such as silicon carbide or graphite coated with silicon carbide. A lower liner assembly **140** formed of silicon carbide can have a thermal conductivity of 200 W/m.K at 20° C. A lower liner assembly **140** formed of graphite can have a thermal conductivity of 107 W/m.K at 20° C. In some embodiments, the thermal conductivity of the lower liner assembly **140** is at least 50% greater, such as at least 100% greater, such as at least five times greater, such as at least ten times greater, such as at least 50 times greater or 100 times greater than the thermal conductivity of the upper liner assembly **130**.

(24) The gas inlet module **107** includes a gas inlet line **123**. The gas inlet line **123** can extend through the body of the gas inlet module **107** and through a portion of the upper liner assembly **130** above the preheat ring **135** to enter the processing region **151**. Process gases from the process gas source **121** can be provided to the gas inlet **123**. Once entering the processing region **151**, the process gases can flow around the one more of the upper portions **162** of the support ring **160**, so that the gas can flow over the substrate **50** positioned on the susceptor **156**. The process gases provided over the substrate **50** can then react and form a deposition layer on the substrate **50**. Unreacted process gases and any byproducts can then be exhausted from the processing region **151** by the exhaust pump **129**. Cleaning gases can be provided to the interior volume **115** through the gas inlet **123** when the substrate support assembly **150** is in a cleaning position **150.sub.2** as described below in reference to FIG. **1C**.

(25) The gas outlet module **108** includes a gas outlet line **128**. The gas outlet line **128** can include an upper portion **128U** extending through the body of the gas outlet module **108** and through a portion of the upper liner assembly **130** above the preheat ring **135**, so that gases can be exhausted from the processing region **151**. The gas outlet line **128** can further include a lower portion **128L** extending through the body of the gas outlet module **108** and through a portion of the lower liner assembly **140**, so that purge gases or other gases (e.g., chilled gases) can be exhausted from the interior volume **115**. Use of the lower portion **128L** located below the processing region **151** reduces the amount of purge gas that reaches processing region **151**, which can prevent the process gases above the substrate **50** from becoming diluted.

(26) The gas inlet module **107** can further include a purge gas inlet line **125**. The purge gas inlet line **125** can extend through the body of the gas inlet module **107** and through a portion of the lower liner assembly **140** to enter the interior volume **115** below the substrate support assembly **150**

when the substrate support assembly **150** is in the processing position **150.sub.1** shown in FIG. **1A**. The purge gas can prevent process gases from entering the portions of the interior volume **115** below the substrate support assembly **150** when the substrate support assembly **150** is in the processing position **150.sub.1**. Some of the purge gases can flow up around the lower plate **153** and between small gaps between the lower plate **153** and the bottom of the upper liner assembly **130**. The purge gas can be exhausted by the exhaust pump **129** from the interior volume **115** in a similar manner as described above for the process gases. Also, most of the purge gas can be exhausted through the lower portion **128L** of the outlet line **128** to reduce the amount of purge gas that enters the processing region **151**.

(27) The process chamber **101** can further include a chilled gas line **127** extending from the chilled gas source **126**, through the actuator **147**, the shaft **146**, and to a gas inlet **154** included in the lower plate **153**, so that the chilled gas from the chilled gas source **126** can enter the interior volume **115** of the process chamber **101**. The chilled gas can be used to cool the substrate **50** when the substrate support assembly **150** is in the cooling position **150.sub.2** as shown in FIG. **1C** after a process is performed on the substrate **50**. In some embodiments, the chilled gases can be the same as the purge gases. For example, in some embodiments, the purge gas and the chilled gas can include one or more of nitrogen, argon, and helium.

(28) The processing system **100** also includes the controller **175** for controlling processes performed by the processing system **100**. The controller **175** can be any type of controller used in an industrial setting, such as a programmable logic controller (PLC). The controller **175** includes a processor **177**, a memory **176**, and input/output (I/O) circuits **178**. The controller **175** can further include one or more of the following components (not shown), such as one or more power supplies, clocks, communication components (e.g., network interface card), and user interfaces typically found in controllers for semiconductor equipment.

(29) The memory **176** can include non-transitory memory. The non-transitory memory can be used to store the programs and settings described below. The memory **176** can include one or more readily available types of memory, such as read only memory (ROM) (e.g., electrically erasable programmable read-only memory (EEPROM), flash memory, floppy disk, hard disk, or random access memory (RAM) (e.g., non-volatile random access memory (NVRAM)).

(30) The processor **177** is configured to execute various programs stored in the memory **176**, such as epitaxial deposition processes, cleaning processes and sequences for moving the substrate support assembly **150** and exchanging substrates as described in further detail below. During execution of these programs, the controller **175** can communicate to I/O devices through the I/O circuits **178**. For example, during execution of these programs and communication through the I/O circuits **178**, the controller **175** can move the substrate support assembly **150** to the processing position in FIG. **1A** or to the cleaning position shown in FIG. **1C**. The memory **176** can further include various operational settings used to control the processes performed in the process chamber **101**, such as the temperature, pressure, and gas flow rates during a deposition, cleaning, or cooling procedure.

(31) FIG. **1B** is a close-up cross-sectional view of section **1B** of FIG. **1A** showing additional detail of the substrate support assembly **150** and liner assemblies **130**, **140**, according to one embodiment.

(32) The upper liner assembly **130** includes an upper portion **131**, a lower portion **132**, and a middle portion **133**. The middle portion **133** includes the ledge **134** that extends inwardly and supports the preheat ring **135**. Although the portions **131-133** are shown as separate components in some embodiments, the portions **131-133** can be formed of one integral body. In some embodiments, the portions **131-133** of the upper liner assembly **130** can be formed of a thermally insulating material (i.e., material having low thermal conductivity), such as quartz or silicon-impregnated quartz. The thermal insulation properties of the portions **131-133** can help the upper portion of the process chamber **101** retain heat, so that heating the upper portion of the interior volume **115** during processes (e.g., depositions) and cleaning procedures is accomplished more

quickly and with less energy.

(33) The lower liner assembly **140** can include an upper portion **141** and a lower portion **142**. The lower portion **142** can include a ledge **143** that is configured to support the lower plate **153** of the substrate support assembly **150** when the substrate support assembly **150** is in the cleaning position as shown in FIG. **1C**. In some embodiments, the lower liner assembly **140** can be formed of a thermally conductive material, such as graphite coated with silicon carbide.

(34) During operation, the purge gas **P** can enter the interior volume **115** through the gas inlet **125**. Although most of purge gas **P** can be exhausted through the lower portion **128L** of the gas outlet line **128** (see FIG. **1A**), some of the purge gas **P** can flow up into the processing volume **151**. A pressure $P_{sub.1}$ in the processing volume **151** can be configured to be lower than a pressure $P_{sub.2}$ in the portion of the interior volume **115** below the processing volume **151**. Due to this pressure differential, some of the purge gas **P** flows up into the processing volume **151** through a small gap between the top of the lower plate **153** and the bottom of the lower portion **132** of the upper liner assembly **130** as well through a small gap between the top of the central ringed portion **163** and the bottom of the preheat ring **135**. These small gaps can be from about 0.1 mm to about 2.0 mm, such as from about 0.4 mm to about 1.0 mm. The higher pressure $P_{sub.2}$ in the interior volume **115** below the lower plate **153** compared to the pressure $P_{sub.1}$ in the processing volume **151** prevents process gases from entering the portion of processing volume **115** below the lower plate **153**, which prevents unintended depositions on components below the lower plate **153**.

(35) In some embodiments (not shown), the central ringed portion **163** only extends inwardly relative to the lower portion **161** and the upper portion **162**. This can allow the upper plate **168** to have a same gap (e.g., a few mm) between the upper portion **131** of the upper liner assembly **130** when the substrate support assembly **150** is in the processing position **150.sub.1** shown in FIGS. **1A** and **1B** (see FIGS. **1A** and **1B**) as the gap between the upper plate **168** and the lower portion **132** of the upper liner assembly **130** when the substrate support assembly **150** is in the cleaning position **150.sub.2** as shown in FIG. **1C**.

(36) FIG. **1C** is a cross-sectional view of the processing system **100** with the substrate support assembly **150** in a cleaning position **150.sub.2**, according to one embodiment. In FIG. **1C**, the substrate support assembly **150** has been lowered relative to the processing position **150.sub.1** shown in FIG. **1A**. With the substrate support assembly **150** in the cleaning position **150.sub.2**, components in a cleaning region **159** of the interior volume **115** above the upper plate **168** can be cleaned by cleaning gases supplied by the cleaning gas source **122**. The cleaning region **159** can be substantially bounded by the upper plate **168** and the lower portion **132** of the upper liner assembly **130** on the bottom, the upper chamber **111** on the top, and the upper liner assembly **130** around the sides. In FIG. **1C**, the substrate **50** is located in a cooling region **152** below the upper plate **168**. The cooling region **152** is substantially bounded by the upper plate **168** on the top, the lower plate **153** on the bottom, and the lower liner assembly **140** around the sides.

(37) In some embodiments, the lower plate **153** can be positioned on a top of the ledge **143** of the lower liner assembly **140**. In other embodiments, the lower plate **153** can be positioned slightly above the ledge **143**, so that the substrate support assembly **150** can be rotated during the cleaning procedure and cooling process to increase the cooling rate and/or uniformity of cooling for the substrate **50**.

(38) When the substrate support assembly **150** is in the cleaning position **150.sub.2**, cleaning gases from the cleaning gas source **122** can be provided to the cleaning region **159** while purge gases from the purge gas source **124** and/or chilled gases from the chilled gas source **126** can be provided to the cooling region **152**. In some embodiments, cleaning gases, such as hydrogen chloride, can be provided to the cleaning region **159** to clean the components in the cleaning region **159**, such as the upper liner assembly **130**, the preheat ring **135**, and the upper chamber **111**. In some embodiments, the cleaning procedure can be performed to operate the cleaning region **159** (1) at a temperature from about 400° C. to about 1200° C., such as from about 700° C. to about 1000° C. and (2) at a

pressure from about 5 Torr to about 760 Torr, such as about 30 Torr. The process performed on the substrate **50** when the substrate support assembly **150** is in the processing position **150.sub.1** (see FIG. **1A**), can be performed within the same ranges of temperature and pressure described above for the cleaning procedure.

(39) Conversely, the purge gas from the purge gas source **124** and/or the chilled gas from the chilled gas source **126** can be used to operate the cooling region **152** when the substrate support assembly **150** is in the cleaning position **150.sub.2** at a lower temperature and a higher pressure. For example, in some embodiments, the cooling region **152** is cooled to a temperature from about 300° C. to about 800° C., such as a temperature from about 500° C. to about 700° C. In some embodiments, the chilled gas can be provided to the cooling region **152** through the gas inlet **154** at a temperature below room temperature (e.g., <20° C.) or a temperature below 0° C. The cooling region **152** can be operated at a pressure P.sub.4 that is higher (e.g., by a few Torr) than a pressure P.sub.3 of the cleaning region **159** to ensure that cleaning gases (e.g., HCl gas) do not enter the cooling region **159** and interact with the substrate **50**. Due to this pressure differential, some purge gas P and/or chilled gas can enter the cleaning region **159** though the gap between the upper plate **168** and the lower portion **132** of the upper liner assembly **130**. The lower portion **128L** of the gas outlet line **128** can be used to limit the amount of purge gas and/or chilled gas that enters the cleaning region **159** by exhausting the purge gas and/or chilled gas directly from the cooling region **152**.

(40) The process chamber **101** can further include a slit valve **195** or other device configured to make an opening to transfer substrates **50** into and out of the interior volume **115**. The slit valve **195** can be opened to allow access to the cooling region **152** by a transfer robot (not shown) to remove the substrate **50** in the cooling region **152** of the process chamber **101** and then add a new substrate **50** to the cooling region **152** of the process chamber **101**. In some embodiments, this exchange of replacing the substrate **50** in the cooling region **152** with a new substrate **50** can happen while the cleaning region **159** is being cleaned by the cleaning gases. Enabling an exchange of substrates **50** while a portion of the process chamber is being cleaned can significantly increase throughput. Conventional processing systems have required removal of substrates during cleaning procedures, which increases the average amount of time to perform a given process on a substrate in the process chamber that can only be cleaned when there are no substrates in the process chamber. Furthermore, for many high-temperature processes, the substrate is cooled before the substrate is removed. Thus, for many conventional processes, such as an epitaxial deposition, (1) the process is performed, then (2) the substrate is cooled, then (3) the substrate is removed, then (4) the cleaning procedure is performed with no substrate in the process chamber, and then (5) the new substrate is positioned in the process chamber after the cleaning procedure is completed. The improvements offered by this disclosure allow operations (2), (3), and (5) all to be performed while the cleaning procedure (i.e., operation (4)) is being performed.

(41) Furthermore, because the cleaning region **159** is substantially isolated from the cooling region **152**, these regions can be configured to be simultaneously operated at significantly different temperatures, such as temperatures that differ by at least 50° C., such as by at least 100° C., such as by at least 300° C. or more. This can allow the temperature of components in the cleaning region **159** to remain closer to temperatures used during the process (e.g., epitaxial deposition), which can allow target process temperatures to be reached more quickly and with less energy when the new substrate **50** is positioned in the processing position **150.sub.1** as shown in FIG. **1A**, which can also increase throughput for the process chamber **101** and decrease costs. Additionally, forming the upper liner assembly **130** of a material having low thermal conductivity enables the upper liner assembly to retain heat while forming the lower liner assembly **140** of a material having high thermal conductivity enables the temperature of the cooling region **152** to rapidly cool, so that the substrate **50** can be cooled and removed from the process chamber **101** more quickly.

(42) FIG. **2A** is a cross-sectional view of a processing system **200** with a substrate support

assembly **250** in a process position **250.sub.1**, according to one embodiment. FIG. 2B is a cross-sectional view of the processing system **200** with the substrate support assembly **250** in a cleaning position **250.sub.2**, according to one embodiment.

(43) In FIG. 2A, the substrate **50** is positioned in a processing region **251**, which is generally similar to the processing region **151** described above. The temperature and pressure in the processing region **251** and other regions of the interior volume **115** can be operated within the same ranges described above for the process chamber **101** when the substrate support assembly is in the processing position **250.sub.1**. For example, FIG. 2A shows the same pressures as shown in FIG. 1B with the pressure $P_{sub.1}$ in the processing region **251** being less than the pressure $P_{sub.2}$ below the lower plate **153** of the substrate support assembly **250**.

(44) In FIG. 2B, the substrate **50** is positioned in a cooling region **252**, which is generally similar to the cooling region **152** described above. The region above the upper plate **168** in FIG. 2B is referred to as the cleaning region **259**, which is generally similar to the cleaning region **159** described above. When the substrate support assembly **250** is in the cleaning position **250.sub.2**, the temperature and pressure of the cooling region **252** and the cleaning region **259** can be operated within the same ranges as those ranges described above for the cooling region **152** and the cleaning region **159** in reference to FIG. 1C. For example, FIG. 2B shows the same pressures as shown in FIG. 1C with the pressure $P_{sub.3}$ in the cleaning region **259** being less than the pressure $P_{sub.4}$ in the cooling region **252**.

(45) The processing system **200** is the same as the processing system **100** described above except that the processing system **200** includes a process chamber **201** instead of the process chamber **101** described above. The process chamber **201** is the same as the process chamber **101** described above except for the following differences. The process chamber **201** includes a lower liner assembly **240** instead of the lower liner assembly **140** described above. Furthermore, the process chamber **201** includes a substrate support apparatus **245** instead of the substrate support apparatus **145** described above. Some dimensions of the upper liner assembly **130** are also adjusted but the overall design is the same, so the same reference numerals are used to describe the upper liner assembly **130** in reference to FIGS. 2A and 2B.

(46) In the process chamber **201**, the lower liner assembly **240** extends closer to the exterior of the process chamber than the lower liner assembly **140** described above. In some embodiments, the lower liner assembly **240** can extend to the exterior of the process chamber **201**. The lower liner assembly **240** can be formed of a material having high thermal conductivity, such as the materials described above for forming the lower liner assembly **140**. Having the lower liner assembly **240** extend closer to the exterior of the process chamber **201** compared to the liner assembly **140** (see FIG. 1A) or to the exterior of the process chamber **201** can allow the lower liner assembly **240** to lose heat at a more rapid rate when the substrate **50** is being cooled. This increased rate of cooling can allow the substrate **50** to cool more rapidly and can allow a cooled substrate **50** to be removed from the process chamber **201** after a shorter time period compared to longer cooling processes, which can increase throughput of the process chamber in some embodiments.

(47) The substrate support apparatus **245** is the same as the substrate support apparatus **145** described above except that the substrate support apparatus **245** includes a substrate support assembly **250** instead of the substrate support assembly **150** included in the substrate support apparatus **145**. The substrate support assembly **250** is the same as the substrate support assembly **150** except that the substrate support assembly **250** includes a plurality of supports **260** (e.g., three supports) that support the upper plate **168** instead of the support ring **160** describe above.

(48) Although the supports **260** lack the central ringed portion **163** included in the support ring **160** (FIG. 1A), the supports **260** allow the upper plate **168** to be positioned at a same or highly similar small gap (1) from the upper portion **131** of the liner assembly **130** when the substrate support assembly **250** is in the processing position **250.sub.1** (FIG. 2A) and (2) from the lower portion **132** of the upper liner assembly **130** when the substrate support assembly **250** is in the cleaning position

250.sub.2 (FIG. 2B). This same of highly similar small gap can be from about 0.1 mm to about 2.0 mm, such as from about 0.4 mm to about 1.0 mm. This small gap in both positions **250.sub.1**, **250.sub.2** can (1) reduce the amount of process gas that goes above the upper plate **168** when the substrate support assembly **250** is in the processing position **250.sub.1** (FIG. 2A), which can reduce cleaning, and (2) reduce the amount of purge gas and/or chilled gas that goes above the upper plate **168** when the substrate support assembly **250** is in the cleaning position **250.sub.2** (FIG. 2B), which can prevent these gases from cooling the cleaning region **259** and from diluting the cleaning gases in the cleaning region **259**.

(49) FIG. 3A is a cross-sectional view of a processing system **300** with a substrate support assembly **350** in a process position **350.sub.1**, according to one embodiment. FIG. 3B is a cross-sectional view of the processing system **300** with the substrate support assembly **350** in an intermediate position **350.sub.2**, according to one embodiment. FIG. 3C is a cross-sectional view of the processing system **300** with the substrate support assembly **350** in a cleaning position **350.sub.3**, according to one embodiment.

(50) In FIG. 3A, the substrate **50** is positioned in a processing region **351**, which is generally similar to the processing region **151** described above. The temperature and pressure in the processing region **351** and other regions of the interior volume **115** can be operated within the same ranges described above for the process chamber **101** when the substrate support assembly is in the processing position **350.sub.1**. For example, FIG. 3A shows the same pressures as shown in FIG. 1B with the pressure $P_{sub.1}$ in the processing region **351** being less than the pressure $P_{sub.2}$ below the lower plate **153** of the substrate support assembly **350**.

(51) In FIG. 3C, the substrate **50** is positioned in a cooling region **352**, which is generally similar to the cooling region **152** described above. The region above the upper plate **168** in FIG. 3B is referred to as the cleaning region **359**, which is generally similar to the cleaning region **159** described above. When the substrate support assembly **350** is in the cleaning position **350.sub.3**, the temperature and pressure of the cooling region **352** and the cleaning region **359** can be operated within the same ranges as those ranges described above for the cooling region **152** and the cleaning region **159** in reference to FIG. 1C. For example, FIG. 3C shows the same pressures as shown in FIG. 1C with the pressure $P_{sub.3}$ in the cleaning region **359** being less than the pressure $P_{sub.4}$ in the cooling region **352**.

(52) The processing system **300** is the same as the processing system **200** (see FIG. 2A) described above except that the processing system **300** includes a process chamber **301** instead of the process chamber **201** described above. The process chamber **301** is the same as the process chamber **201** described above except for the following differences. The process chamber **301** includes a preheat ring **335** instead of the preheat ring **135** described above. The process chamber **301** includes an upper liner assembly **330** instead of the upper liner assembly **130** described above. Furthermore, the process chamber **301** includes a substrate support apparatus **345** instead of the substrate support apparatus **245** described above.

(53) The preheat ring **335** is the same as the preheat ring **135** described above except that preheat ring **335** includes a body **336** and an inner extension **337**. The body **336** can be substantially the same (e.g., same or similar material and dimensions) as the preheat ring **135** described above. The inner extension **337** can be formed of the same material as the preheat ring **135** described above. The inner extension **337** can extend downwardly from the inner edge of the body **336**.

(54) The upper liner assembly **330** is the same as the upper liner assembly **130** described above except that the upper liner assembly **330** does not include the lower portion **132** described above and the dimensions of the other portions (e.g., portions **131**, **133**) may be different.

(55) The substrate support apparatus **345** is the same as the substrate support apparatus **245** except that the substrate support apparatus **345** includes a substrate support assembly **350** instead of the substrate support assembly **250** described above. The substrate support assembly **350** is the same as the substrate support assembly **250** described above except that the substrate support assembly **350**

includes a plurality of supports **360** (e.g., three supports **360**) to support the upper plate **168** over the susceptor **156** instead of the supports **260**, and the substrate support assembly **350** further includes an outer ring **370** extending upwardly from the lower plate **153**. The supports **360** can be suspended from the bottom of the upper plate **168** as more clearly shown in FIGS. **3B** and **3C**. In some embodiments, the supports **360** can have a pin shape. The supports **360** can be positioned to contact the susceptor **156** as the susceptor **156** is raised to the processing position **350.sub.1** shown in FIG. **3A**.

(56) The inner extension **337** of the preheat ring **335** and the outer ring **370** of the substrate support assembly **350** are positioned to form a tortuous path for gas (e.g., purge gas) to flow from below the lower plate **153** to a location above the lower plate **153** and directly above or below the susceptor **156** in the processing region **351** when the substrate support assembly **350** is in the processing position **350.sub.1**. This tortuous path with no direct line of sight from below the lower plate **153** to a portion of the processing region **351** directly underlying or overlying the susceptor **156** can help reduce the amount of purge gas that enters the processing region **351** and can reduce the likelihood or degree that the purge gas may dilute the process gas that interacts with the substrate **50**.

(57) In FIG. **3B**, as the substrate support assembly **350** is lowered by the actuator **147** to the intermediate position **350.sub.2**, the lower surface of the upper plate **168** contacts the upper surface of the body **336** of the preheat ring **335**. In FIG. **3C**, the substrate support assembly **350** is lowered by the actuator **147** to the cleaning position **350.sub.3**. In the cleaning position **350.sub.3**, the upper plate **168** is resting on the preheat ring **335**, which can create a seal between the cleaning region **359** and the cooling region **352** below the upper plate **168**.

(58) With reference to FIG. **3C**, the process chamber **301** can further include a slit valve **395** or other device configured to make an opening to transfer substrates **50** into and out of the cooling region **352** of the process chamber **301**. The slit valve **395** can be opened to allow access to the cooling region **352** by a transfer robot (not shown) to remove the substrate **50** in the cooling region **352** of the process chamber **301**, and then add a new substrate **50** to the cooling region **352** of the process chamber **301**. The process chamber **301** can further include a gas inlet **396** positioned adjacent to the slit valve **395**. For example, the slit valve **395** and the gas inlet **396** can be positioned at a same angular location relative to the central axis **C**. The gas inlet **396** can be connected to the purge gas source **124** and/or the chilled gas source **126**. The gas inlet **396** is positioned to direct gas at the substrate **50** when the substrate support assembly **350** is in the cleaning position **350.sub.3** shown in FIG. **3C**. Furthermore, the gas flow from the gas inlet **396** can assist in preventing gas from inside the process chamber **301** flowing out through the opening of the slit valve **395** when substrates are exchanged through the opening of the slit valve **395**. In some embodiments, the pressure of the interior volume **115** of the process chamber **301** can be maintained at a lower pressure than an interior pressure of a transfer chamber (not shown) connected to the slit valve **395** to prevent gas from flowing from the interior of the process chamber **301** to the interior of the transfer chamber. The gas inlet **396** can be included in the other process chambers described in this disclosure as well.

(59) FIG. **4** is a process flow diagram of a method **1000** of processing a substrate in a process chamber and cleaning the process chamber, according to one embodiment. The method **1000** is described in reference to FIGS. **1A-3C**. The controller **175** can execute a program stored in memory to perform the method **1000**. The method begins at block **1002**.

(60) At block **1002** a process (e.g., a deposition) is performed on a substrate **50** that is positioned on one of the substrate support assemblies **150**, **250**, **350** described above. Process gases can be introduced to the corresponding processing region **151**, **251**, **351** and purge gases can be introduced below the lower plate **153** of the corresponding substrate support assembly. The lower plate **153** can be positioned in close proximity (e.g., from about 0.4 mm to about 1.0 mm) from one or more other components (e.g., the lower portion **132** of the upper liner assembly **130** or the preheat ring

335) to reduce the amount of purge gas that enters the corresponding processing region during block **1002**. The upper plate **168** of the corresponding substrate support assembly can be positioned in close proximity (e.g., from about 0.4 mm to about 1.0 mm) from one or more other components (e.g., the upper portion **131** of the upper liner assembly **130**) to reduce the amount of process gases that enter portions of the interior volume **115** above the upper plate **168**, which can reduce unintended depositions in these portions of the interior volume **115**.

(61) At block **1004**, the corresponding substrate support assembly **150**, **250**, **350** is moved (e.g., lowered) to a cleaning position **150.sub.2**, **250.sub.2**, **350.sub.3**, so that a cleaning region **159**, **259**, **359** of the interior volume **115** above the upper plate **168** can be cleaned, and the substrate **50** can be cooled in a cooling region **152**, **252**, **352** located below the upper plate **168**. The upper plate **168** can be positioned in close proximity to one or more other components (e.g., the lower portion **132** of the upper liner assembly **130** as shown in FIGS. **1C** and **2B**) or on top of a component, such as on top of the preheat ring **335** (see FIG. **3C**) to substantially seal the corresponding cleaning region **159**, **259**, **359** from the corresponding cooling region **152**, **252**, **352**. This seal prevents or reduces the amount of purge gas and/or cooling gas that can enter the corresponding cleaning region and prevents or reduces the amount of cleaning gases that can reach the cooling region.

(62) At block **1006**, cleaning gases are introduced to the corresponding cleaning region **159**, **259**, **359**, and purge gas and/or chilled gas are introduced to the corresponding cooling region **152**, **252**, **352** at the same time. Thus, the components in the corresponding cleaning region are cleaned at the same time the substrate **50** in the cooling region of the same process chamber is cooled. Notably, the substrate **50** is uncovered in the cooling region to allow the purge gas and/or chilled gas to flow over the surfaces of the substrate **50**, which enables the substrate **50** to be quickly cooled. In some embodiments, the cleaning gases would damage the substrate **50** if the substrate **50** was exposed to the cleaning gases. The substrate **50** is not exposed to the cleaning gases though because the substrate support assembly is positioned to limit the flow of gas between the cleaning region and the cooling region and because the pressure in the cooling region is slightly higher than the pressure in the cleaning region.

(63) At block **1008**, after the existing substrate **50** is cooled to a specified temperature, the processed substrate **50** is replaced with a new substrate **50** while the cleaning gases continue to be provided to the corresponding cleaning region **159**, **259**, **359**.

(64) In some embodiments, the cleaning can be completed before the cooling of the processed substrate **50** is completed. Enabling the cooling of the processed substrate **50** to be performed simultaneously with the cleaning of another portion of the process chamber can substantially increase throughput for a process chamber. In the embodiments described above, the cooling regions are located below the cleaning regions, but in some embodiments, the components in the process chambers can be rearranged, so that the cleaning region is below the cooling region.

(65) While the foregoing is directed to examples of the present disclosure, other and further examples of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A process chamber comprising: a chamber body enclosing an interior volume; a substrate support assembly in the interior volume, the substrate support assembly comprising a substrate receiving surface, an upper plate, and a lower plate, wherein the upper plate is positioned over the substrate receiving surface, and wherein the substrate receiving surface is positioned over and spaced apart from the lower plate; a first gas inlet line positioned above the lower plate and below the upper plate when the substrate support assembly is in a first position, the first gas inlet line positioned above the lower plate and above the upper plate when the substrate support is in a second position, wherein the vertical position of the lower plate in a first position is lower than the

vertical position of the lower plate in the second position; a first gas outlet line positioned above the lower plate and below the upper plate when the substrate support assembly is in the first position, the first gas outlet line positioned above the lower plate and above the upper plate when the substrate is in the second position; a second gas inlet line positioned below the lower plate and below the upper plate when the substrate support assembly is in the first position; and a second gas outlet line positioned below the lower plate and below the upper plate when the substrate support assembly is in the first position.

2. The process chamber of claim 1, wherein the upper plate and the lower plate each have a larger cross-sectional area in a horizontal plane than a cross-sectional area of the substrate support in a horizontal plane.

3. The process chamber of claim 1, further comprising an upper liner and a lower liner, the upper liner and the lower liner each disposed around an outer portion of the interior volume, wherein the upper liner is formed of a different material than the lower liner.

4. The process chamber of claim 3, wherein the lower liner has a thermal conductivity that is at least 100% greater than the thermal conductivity of the upper liner.

5. The process chamber of claim 4, wherein the lower liner extends to an exterior of the process chamber.

6. The process chamber of claim 3, wherein the substrate support assembly is configured to move from the first position to the second position, wherein the upper plate is positioned from about 0.4 mm to about 1.0 mm from the upper liner in the first position.

7. The process chamber of claim 6, wherein the upper plate is positioned from about 0.4 mm to about 1.0 mm from the lower liner when the substrate support assembly is in the second position.

8. The process chamber of claim 6, wherein the lower plate is positioned from about 0.4 mm to about 1.0 mm from the upper liner when the substrate support assembly is in the first position.

9. The process chamber of claim 6, further comprising a slit valve positioned below the upper plate when the substrate support assembly is in the second position.

10. A process chamber comprising: a chamber body enclosing an interior volume; a substrate support assembly in the interior volume, the substrate support assembly comprising a substrate receiving surface, an upper plate, and a lower plate, the upper plate positioned over the substrate receiving surface, wherein the substrate receiving surface is positioned over and spaced apart from the lower plate; a first ring positioned above the lower plate of the substrate support assembly, wherein a portion of the upper plate of the substrate support assembly extends over the first ring; a first gas inlet line positioned above the lower plate and below the upper plate when the substrate support assembly is in a first position, wherein the first gas inlet line is positioned above the lower plate and above the upper plate when the substrate support assembly is in a second position; and a first gas outlet line positioned above the lower plate and below the upper plate when the substrate support assembly is in the first position, wherein the first gas outlet line is positioned above the lower plate and above the upper plate when the substrate support assembly is in the second position.

11. The process chamber of claim 10, further comprising an outer ring extending upward from the lower plate.

12. The process chamber of claim 11, wherein the first ring includes a body and an inner extension having a ring shape, and the outer ring underlies a portion of the body of the first ring.

13. The process chamber of claim 12, further comprising an upper liner and a lower liner, the upper liner and the lower liner each disposed around an outer portion of the interior volume, wherein the upper liner is formed of a different material than the lower liner.

14. The process chamber of claim 13, wherein the lower liner has a thermal conductivity that is at least 100% greater than the thermal conductivity of the upper liner.

15. The process chamber of claim 14, wherein the lower liner extends to an exterior of the process chamber.

16. The process chamber of claim 13, wherein the substrate support assembly is configured to move from the first position to the second position, wherein the upper plate is positioned from about 0.4 mm to about 1.0 mm from the upper liner in the first position.

17. The process chamber of claim 16, wherein the upper liner is configured to be supported by the first ring when the substrate support assembly is in the second position.

18. The process chamber of claim 17, wherein the upper plate is spaced apart from the substrate support by a greater distance in the second position than in the first position.

19. The process chamber of claim 10, further comprising: a second gas inlet line positioned below the lower plate and below the upper plate when the substrate support assembly is in the first position, wherein the second gas inlet line is positioned above the lower plate and below the upper plate when the substrate support assembly is in the second position; and a second gas outlet line positioned below the lower plate and below the upper plate when the substrate support assembly is in the first position, wherein the second gas outlet line is positioned above the lower plate and below the upper plate when the substrate support assembly is in the second position.
